

Roboten som ser deg

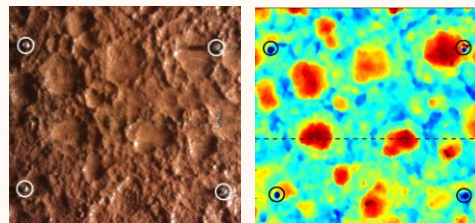
Bildebehandling og ansiktsgjenkjenning – hvordan virker det?

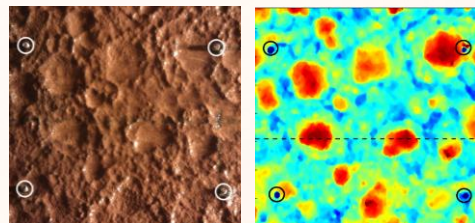
Martin H. Skjelvareid

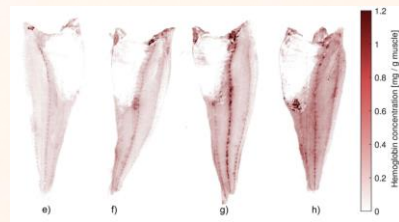
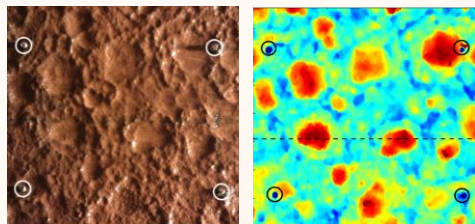












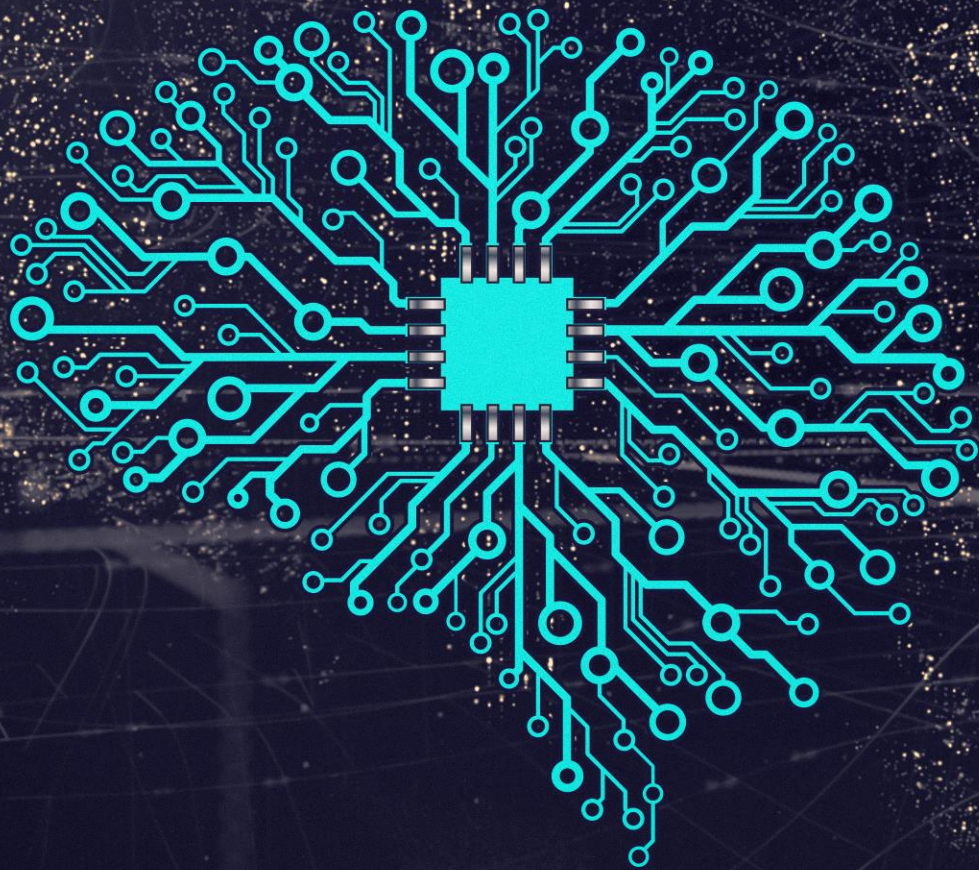


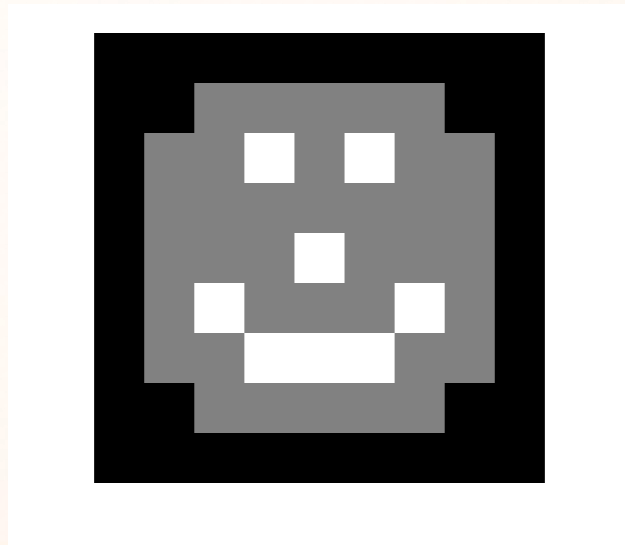
Photo: Mike MacKenzie
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0	0	0	0	0	0	0	0	0
0	0	0.5	0.5	0.5	0.5	0.5	0	0
0	0.5	0.5	1.0	0.5	1.0	0.5	0.5	0
0	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0
0	0.5	0.5	0.5	1.0	0.5	0.5	0.5	0
0	0.5	1.0	0.5	0.5	0.5	1.0	0.5	0
0	0.5	0.5	1.0	1.0	1.0	0.5	0.5	0
0	0	0.5	0.5	0.5	0.5	0.5	0	0
0	0	0	0	0	0	0	0	0

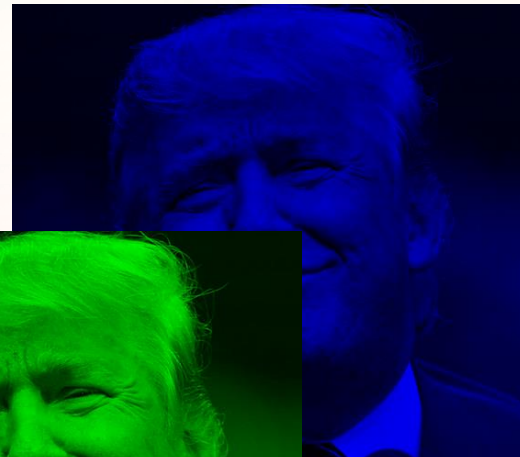
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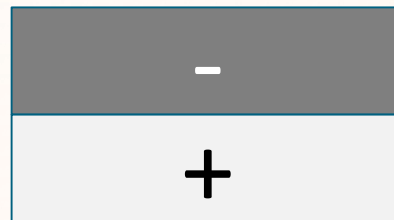






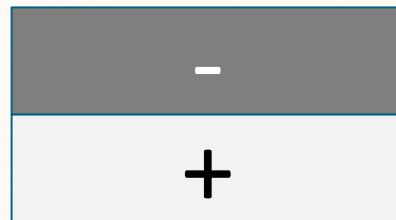
10	4	5	2	9	7
3	11	6	12	1	8

5	2	7	3	1	4
9	12	6	10	8	11



10	4	5	2	9	7
3	11	6	12	1	8

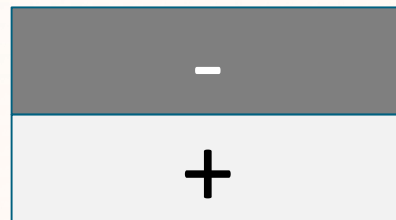
5	2	7	3	1	4
9	12	6	10	8	11



10	4	5	2	9	7
3	11	6	12	1	8

5	2	7	3	1	4
9	12	6	10	8	11

$$= 41 - 37 = 4$$

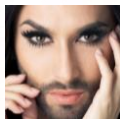
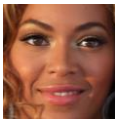


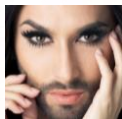
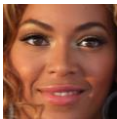
10	4	5	2	9	7
3	11	6	12	1	8

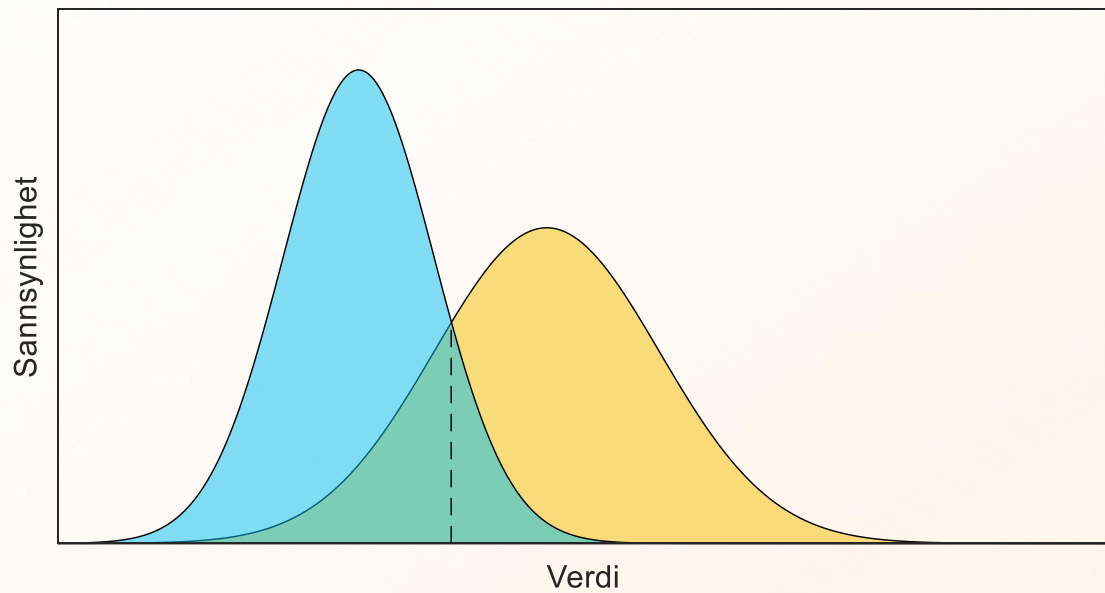
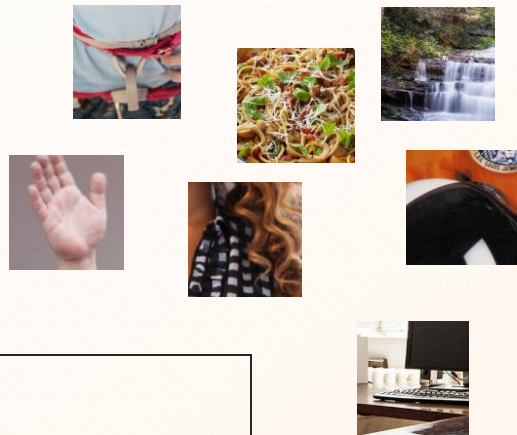
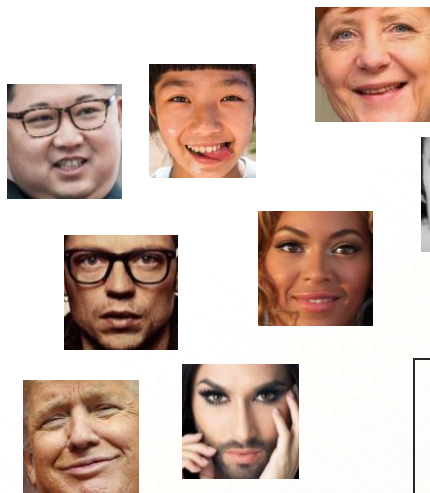
$$= 41 - 37 = 4$$

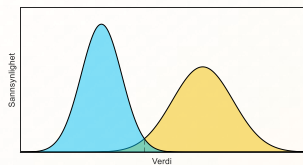
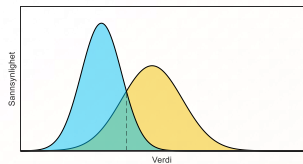
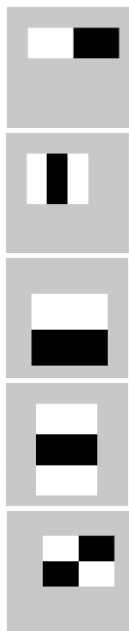
5	2	7	3	1	4
9	12	6	10	8	11

$$= 56 - 22 = 34$$

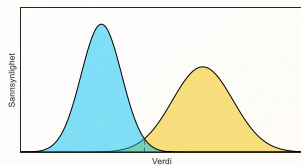
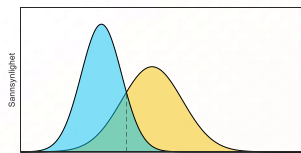
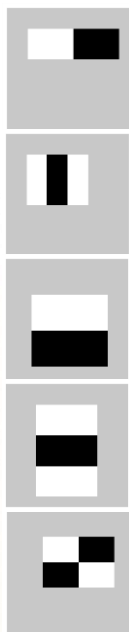






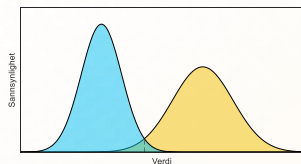
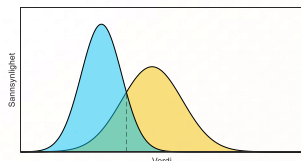
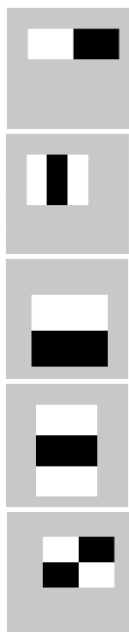


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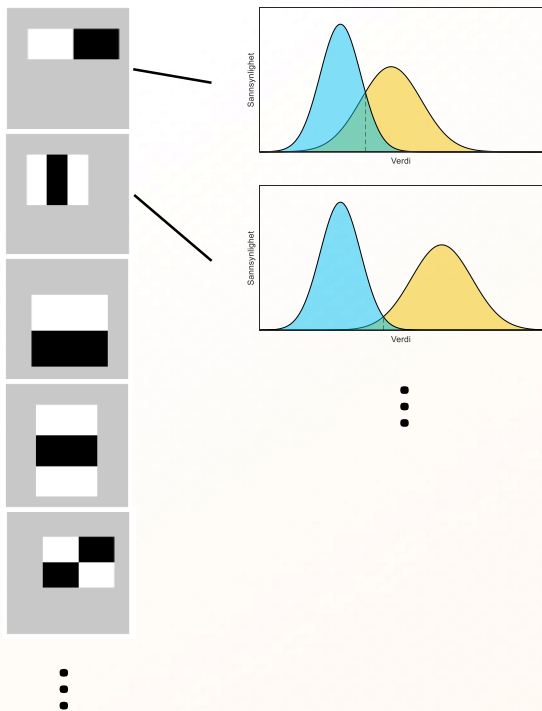
...

for alle *typer* masker:



...

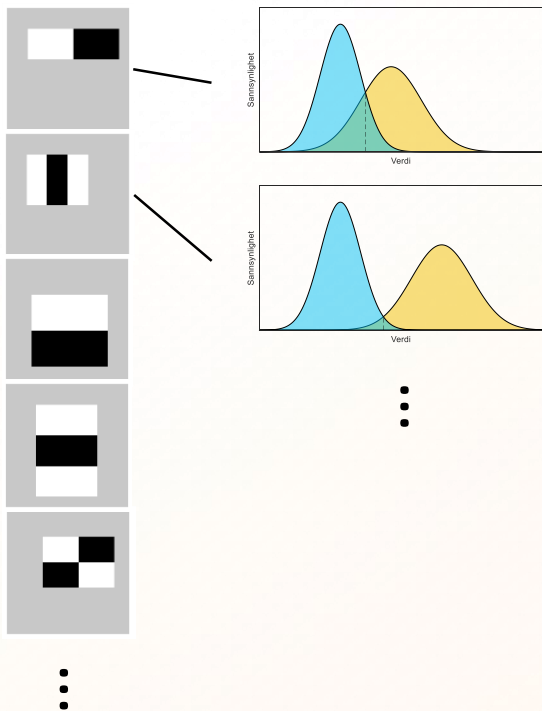
for alle typer masker:
for alle maskestørrelser:



for alle typer masker:

for alle maskestørrelser:

for alle maskeposisjoner:

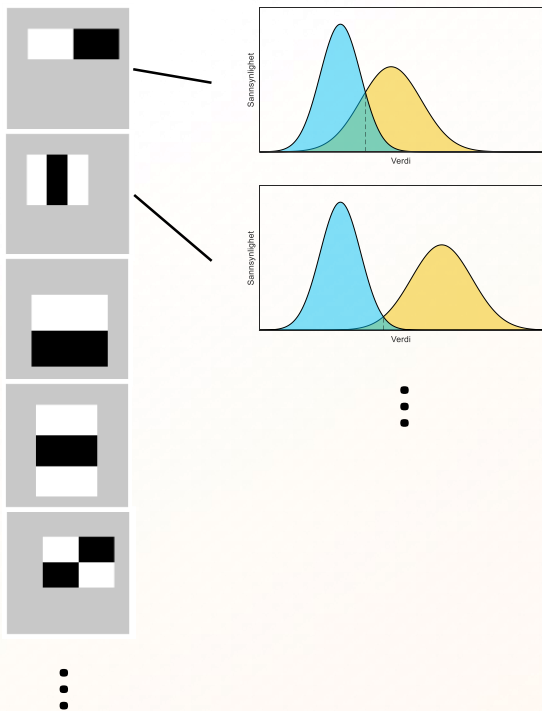


for alle *typer* masker:

for alle maskestørrelser:

for alle maskeposisjoner:

test separasjon *ansikter* vs. *ikke-ansikter*



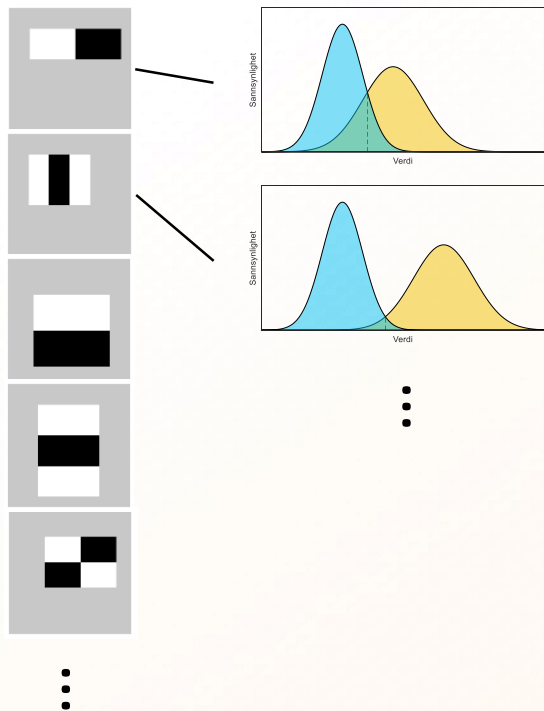
for alle *typer* masker:

for alle maskestørrelser:

for alle maskeposisjoner:

test separasjon *ansikter* vs. *ikke-ansikter*

-> bruk maskene som separerer best



for alle typer masker:

for alle maskestørrelser:

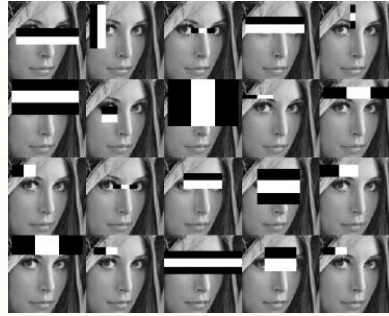
for alle maskeposisjoner:

test separasjon *ansikter* vs. *ikke-ansikter*

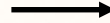
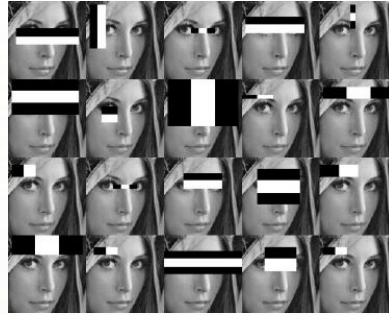
-> bruk maskene som separerer best

*Masker kan kombineres ved «stemming»
«Vekt» etter hvor god separasjon masken gir*

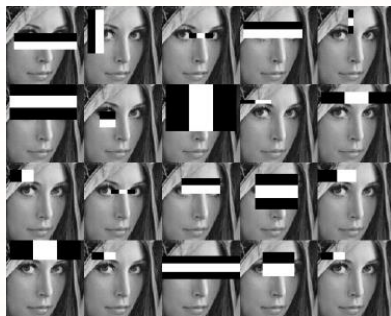




1)



6,7	6,5	2,7	6,9	4,3
7,5	1,7	0,4	3,1	3,8
7,4	7,0	0,9	9,5	7,6
3,9	0,3	8,2	0,3	7,9



6,7	6,5	2,7	6,9	4,3
7,5	1,7	0,4	3,1	3,8
7,4	7,0	0,9	9,5	7,6
3,9	0,3	8,2	0,3	7,9

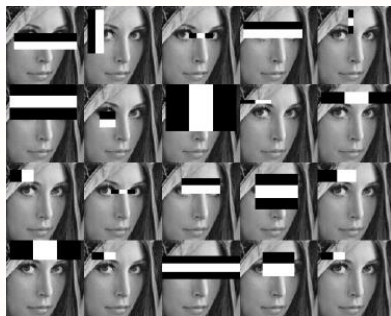
Test 1



Kanskje
et ansikt



Ikke et
ansikt



6,7	6,5	2,7	6,9	4,3
7,5	1,7	0,4	3,1	3,8
7,4	7,0	0,9	9,5	7,6
3,9	0,3	8,2	0,3	7,9

Test 1

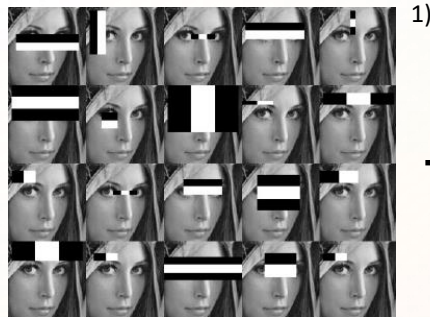
Ikke et
ansikt

Kanskje
et ansikt

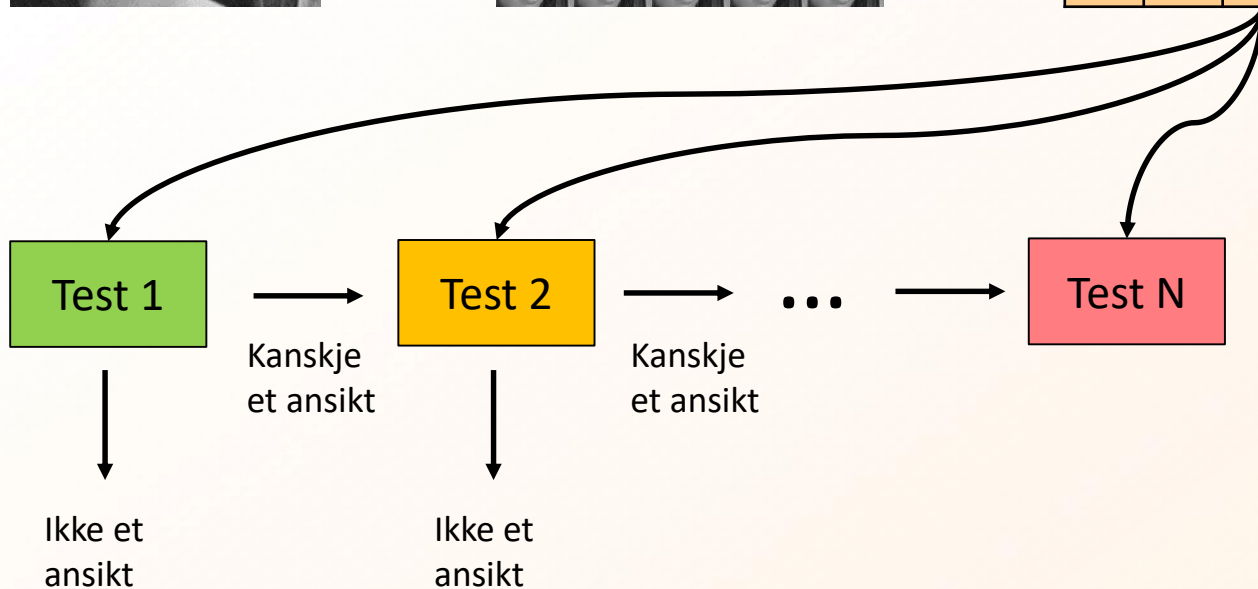
Test 2

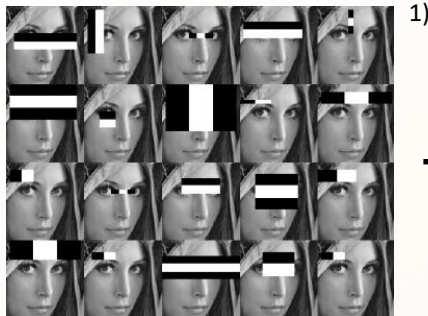
Ikke et
ansikt

Kanskje
et ansikt



6,7	6,5	2,7	6,9	4,3
7,5	1,7	0,4	3,1	3,8
7,4	7,0	0,9	9,5	7,6
3,9	0,3	8,2	0,3	7,9





6,7	6,5	2,7	6,9	4,3
7,5	1,7	0,4	3,1	3,8
7,4	7,0	0,9	9,5	7,6
3,9	0,3	8,2	0,3	7,9

Test 1

Ikke et
ansikt

Kanskje
et ansikt

Test 2

Ikke et
ansikt

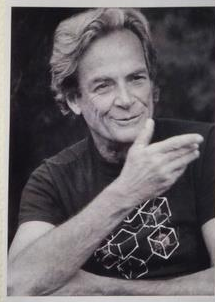
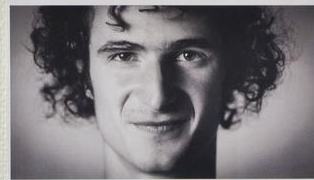
Kanskje
et ansikt

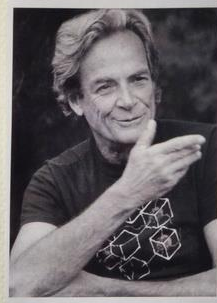
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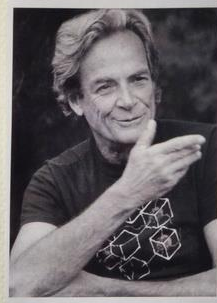
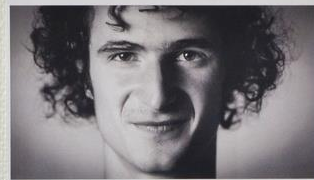
Test N

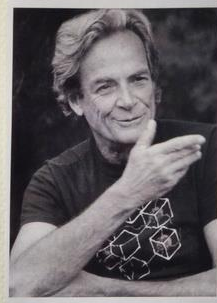
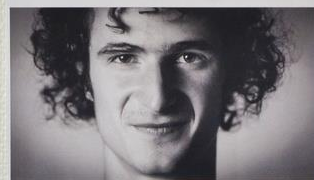
Ikke et
ansikt

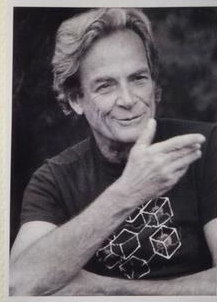
Definitivt
et ansikt!

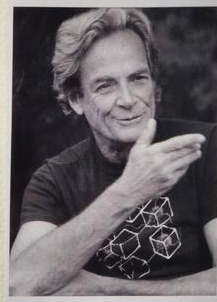


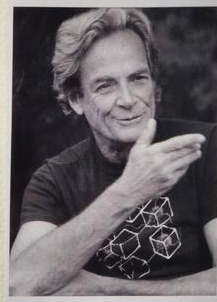
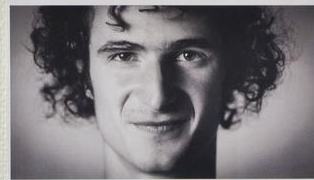


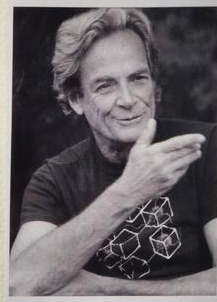


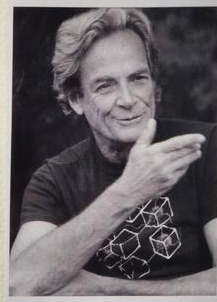
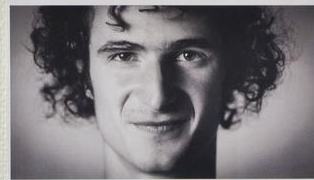


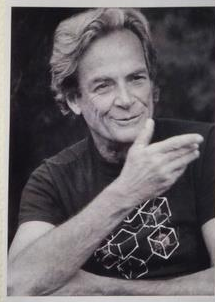
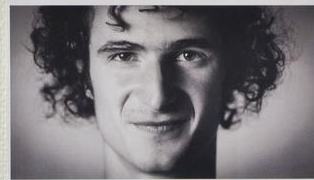


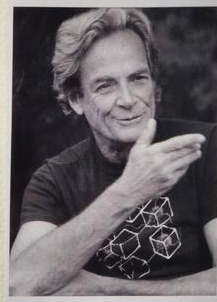
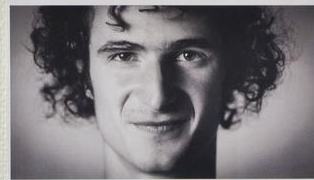


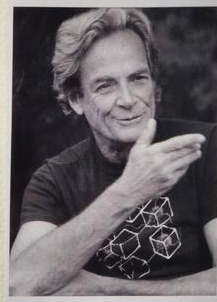


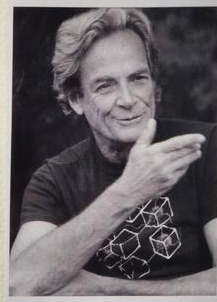
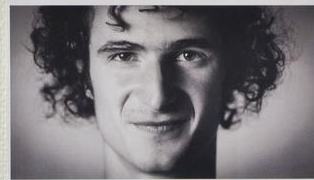
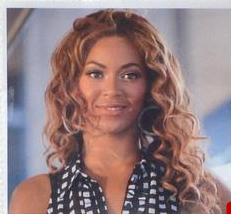


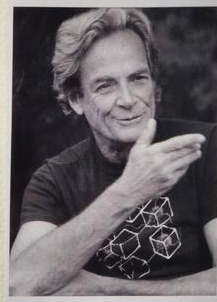
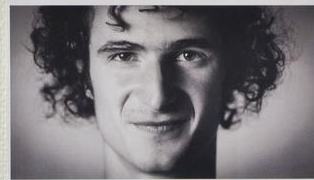


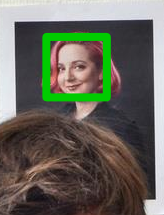
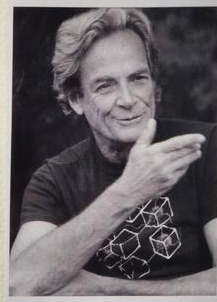
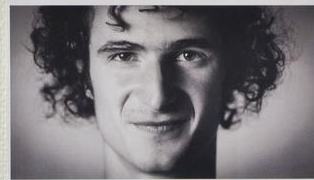


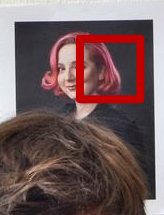
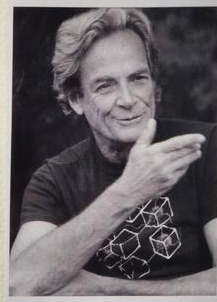
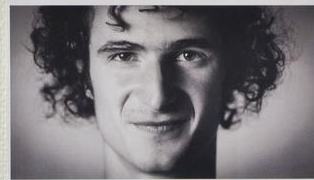


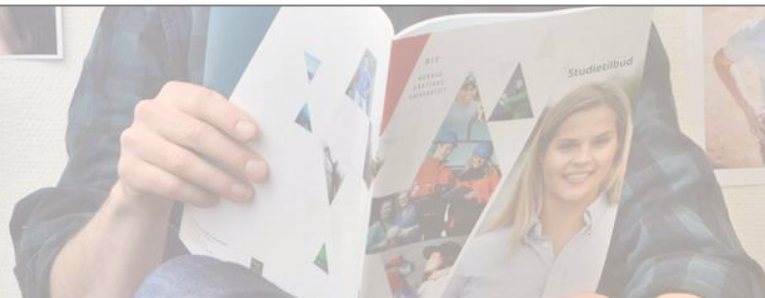
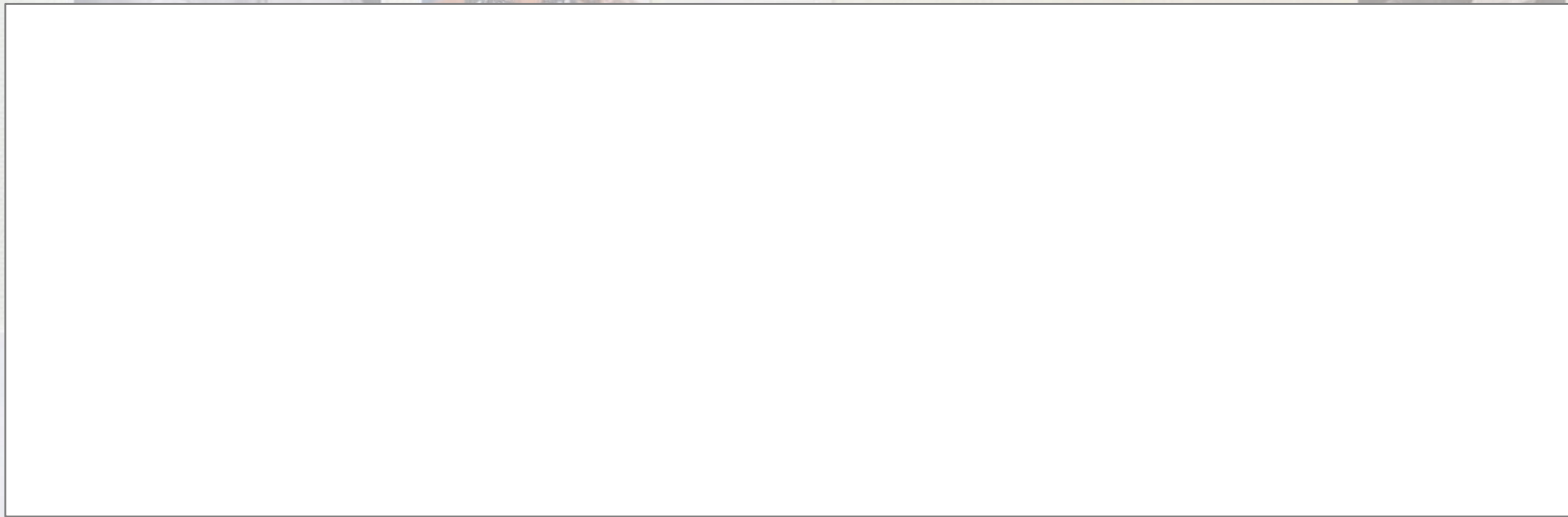
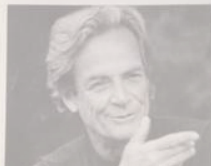






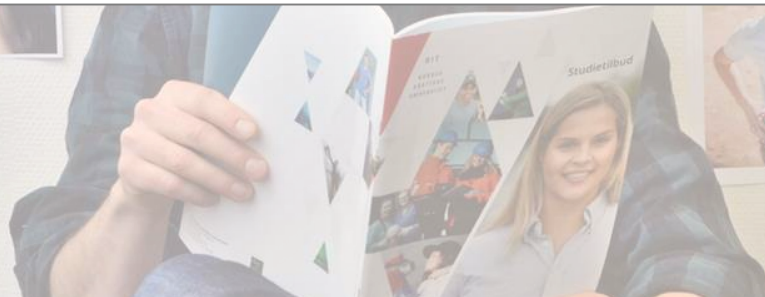


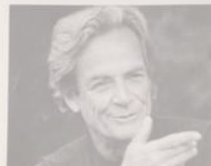




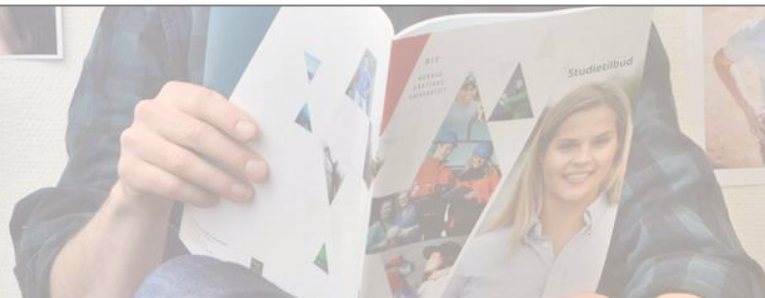


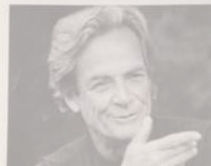
for alle størrelser av utsnitt:



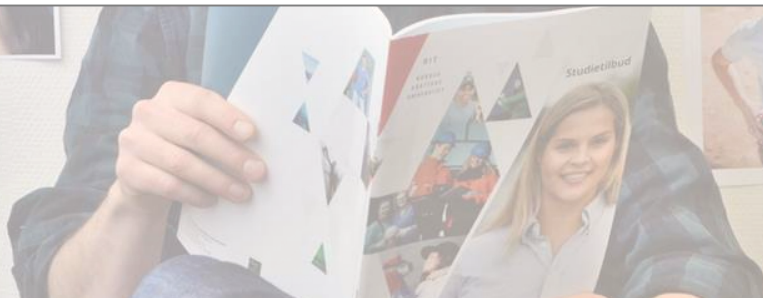


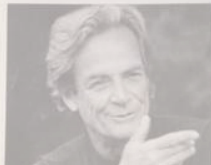
for alle størrelser av utsnitt:
for alle posisjoner i bildet:



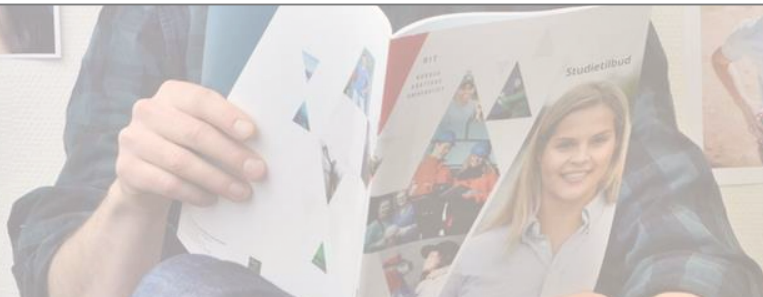


for alle størrelser av utsnitt:
for alle posisjoner i bildet:
kjør ansikts-test



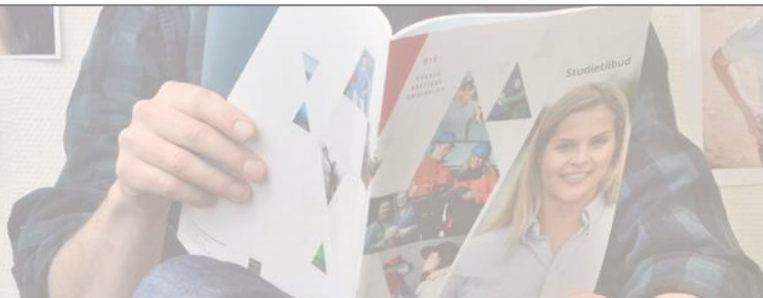


for alle størrelser av utsnitt:
for alle posisjoner i bildet:
kjør ansikts-test
hvis utsnittet er et ansikt:





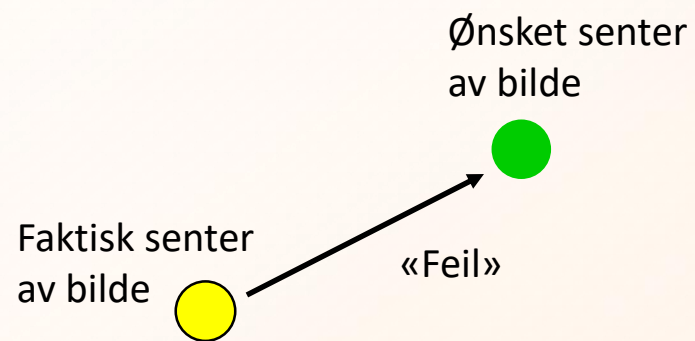
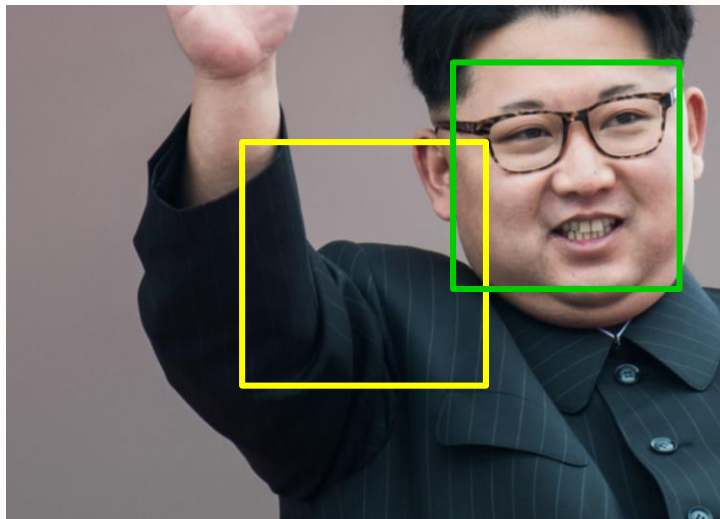
for alle størrelser av utsnitt:
for alle posisjoner i bildet:
kjør ansikts-test
hvis utsnittet er et ansikt:
marker med grønn ramme



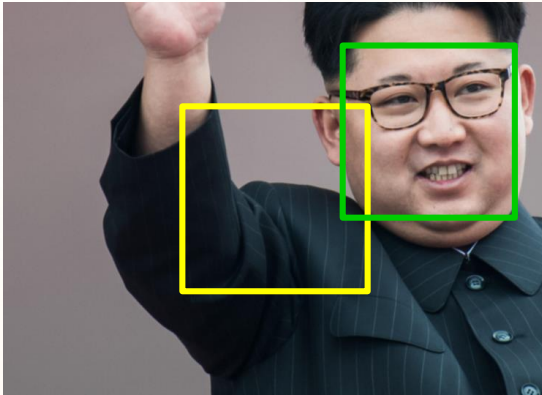


A close-up, low-angle shot of WALL-E's face. His two large, teardrop-shaped eyes are the central focus, each containing a vibrant blue, starry pattern. The eyes are set within dark, metallic frames that show signs of wear and small rivets. Below the eyes, parts of his yellow, segmented body and mechanical joints are visible, though slightly out of focus. The background is dark and textured, suggesting an industrial or space environment.

Men tilbake til roboten...

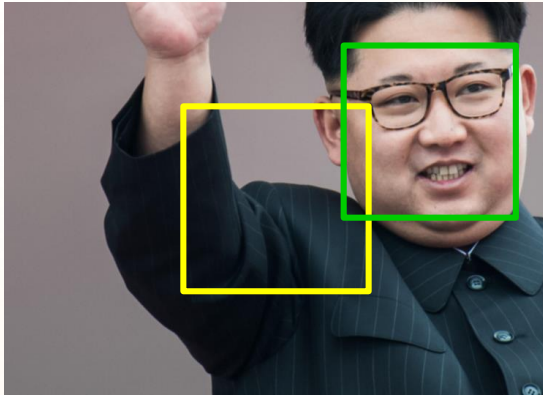


feil = ønsket posisjon – faktisk posisjon



feil = ønsket posisjon – faktisk posisjon
neste kameravinkel = gammel vinkel + feil · kh

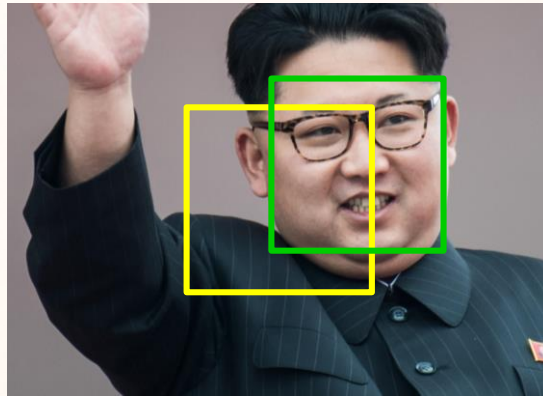
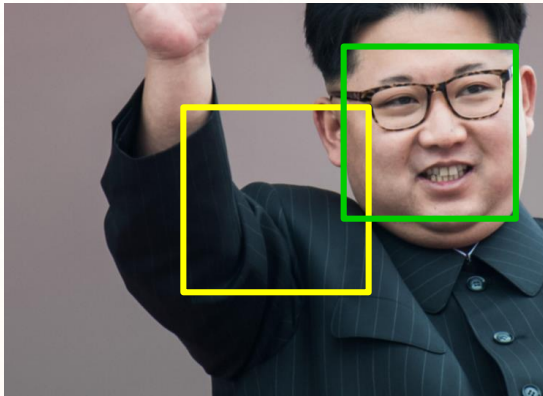
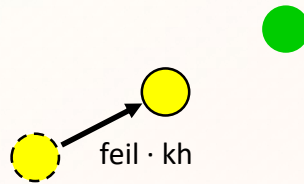
(kh : «korreksjonshastighet»)



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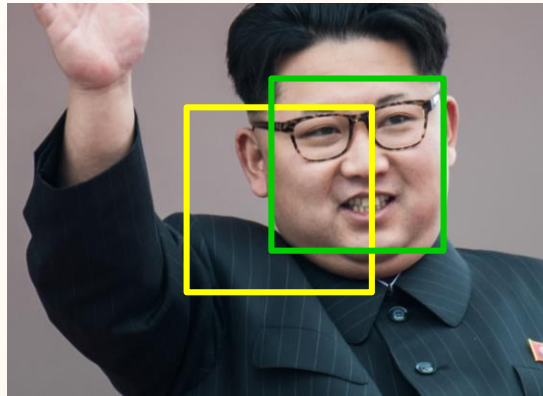
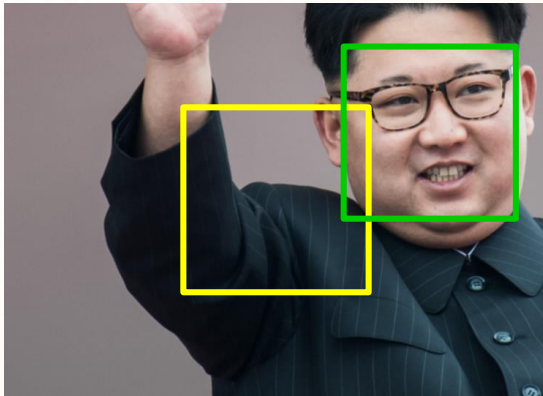
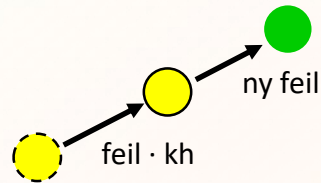
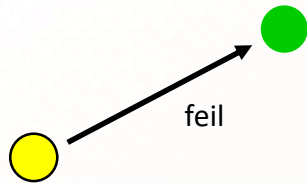
(kh : «korreksjonshastighet»)



feil = ønsket posisjon – faktisk posisjon

neste kameravinkel = gammel vinkel + feil · kh

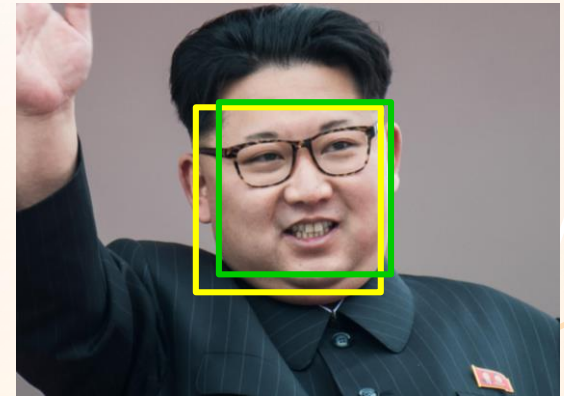
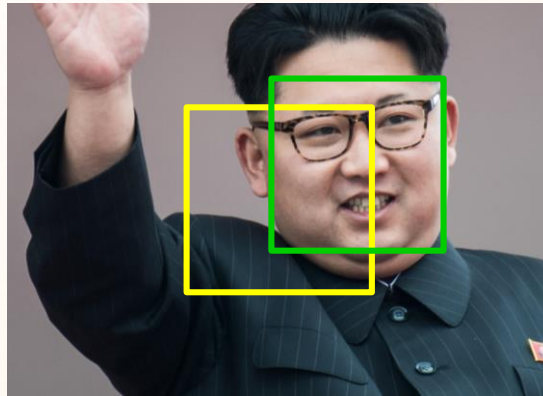
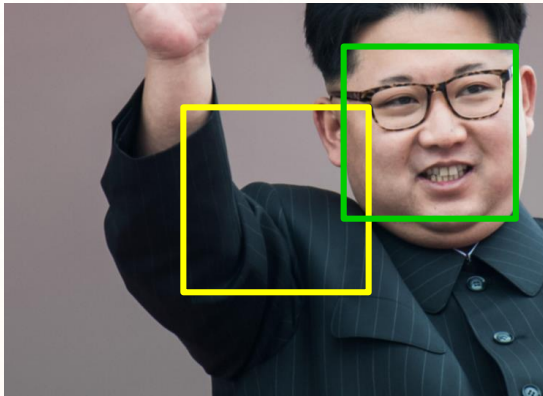
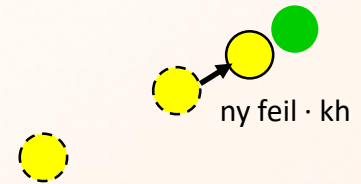
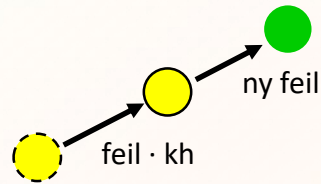
(kh : «korreksjonshastighet»)



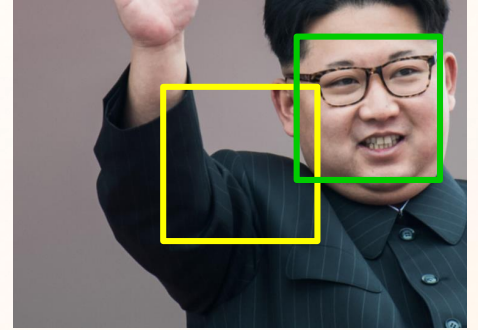
feil = ønsket posisjon – faktisk posisjon

neste kameravinkel = gammel vinkel + feil · kh

(kh : «korreksjonshastighet»)



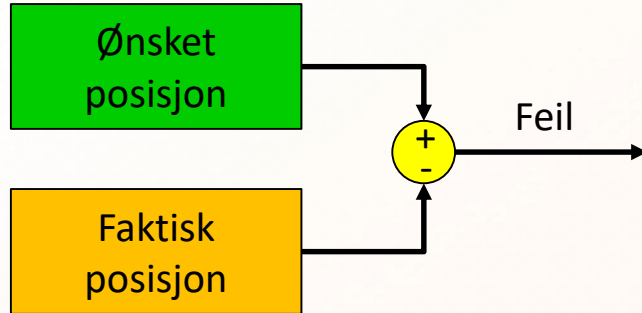
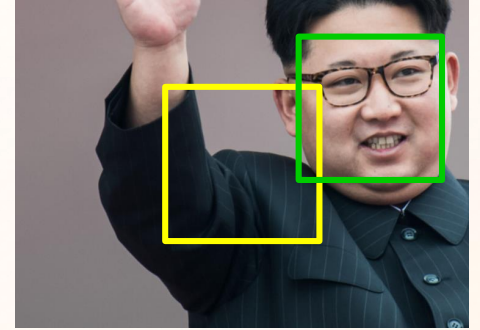
feil = ønsket posisjon – faktisk posisjon



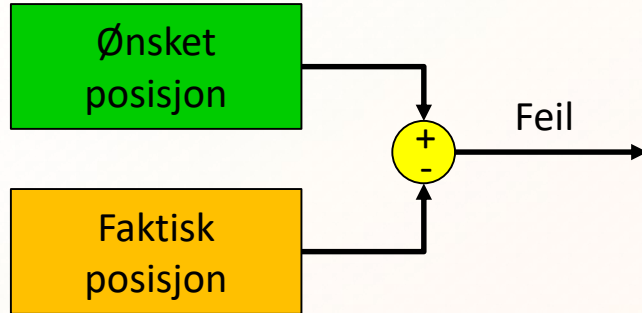
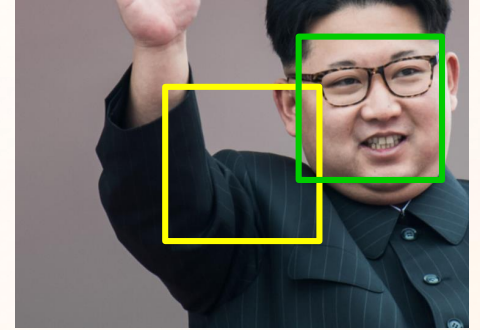
+

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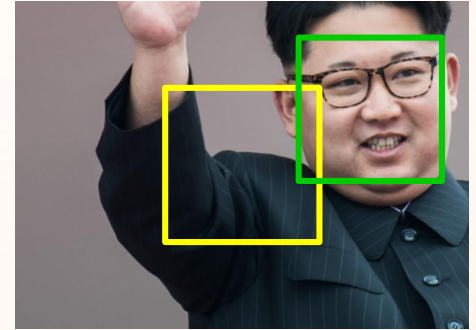
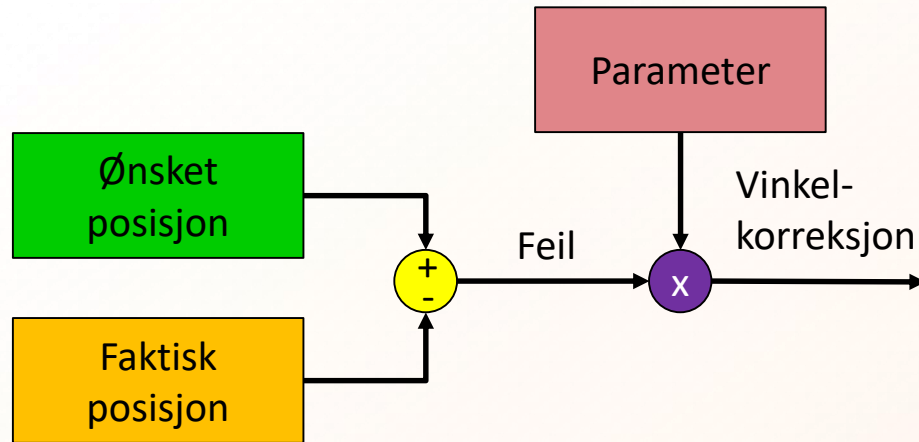
$\text{feil} = \text{ønsket posisjon} - \text{faktisk posisjon}$



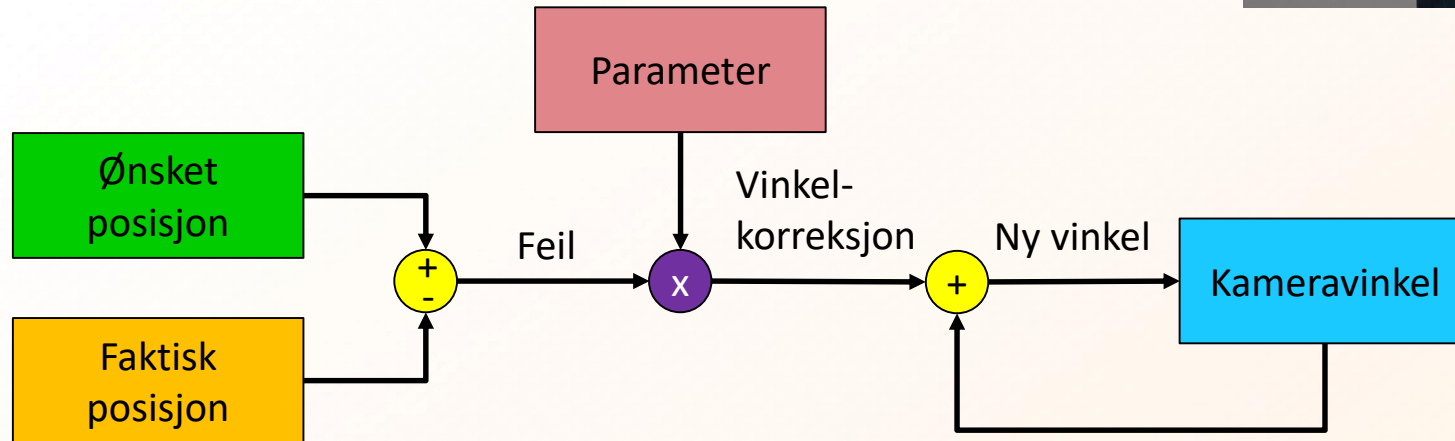
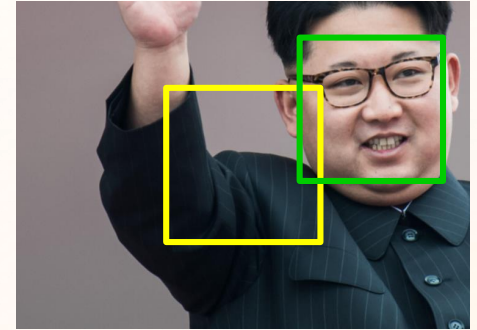
$\text{feil} = \text{\AA nsket posisjon} - \text{faktisk posisjon}$
 $\text{neste kameravinkel} = \text{gammel vinkel} + \text{feil} * \text{parameter}$

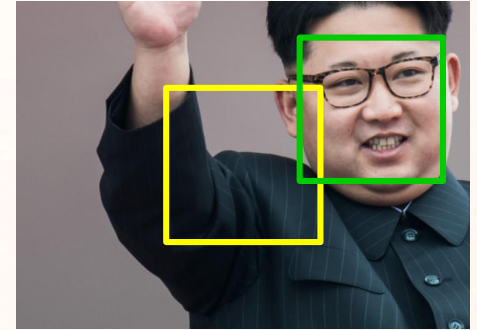


feil = ønsket posisjon – faktisk posisjon
neste kameravinkel = gammel vinkel + feil * parameter

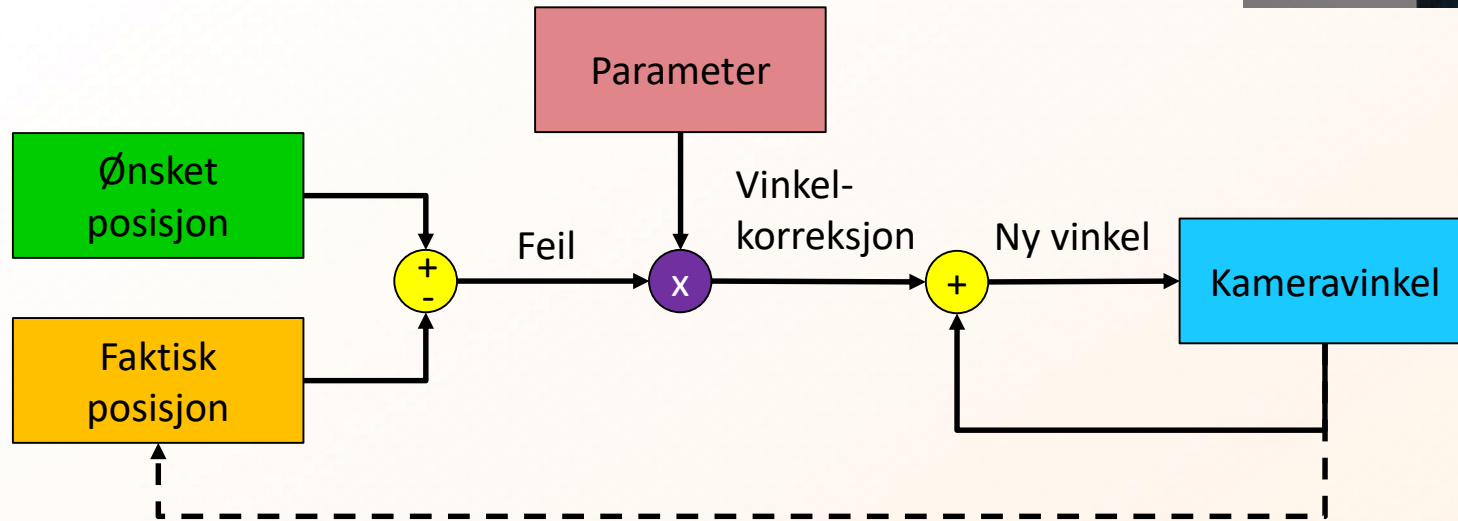


feil = ønsket posisjon – faktisk posisjon
neste kameravinkel = gammel vinkel + feil * parameter





feil = ønsket posisjon – faktisk posisjon
neste kameravinkel = gammel vinkel + feil * parameter



for hvert bilde i videostrøm:

- søk etter ansikter

- velg ansiktet nærmest senter

- beregn «feil» i ansiktsposisjon

- oppdater kameravinkel

- vis detekterte ansikter

for hvert bilde i videostrøm:
søk etter ansikter
velg ansiktet nærmest senter
beregner «feil» i ansiktsposisjon
oppdater kameravinkel
vis detekterte ansikter

```
# %% Main loop
try:
    while True:
        # Capture frame-by-frame, convert to greyscale for face detection
        ret, frame = video_capture.read() # Get image from video
        gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY) # Color to grayscale
        (frameHeight, frameWidth, nChannels) = frame.shape # Get image size

        # Detect faces
        faces = faceCascade.detectMultiScale(
            gray,
            scaleFactor=scaleFactor,
            minNeighbors=minNeighbors,
            minSize=minSize,
            maxSize=maxSize,
            flags=cv2.CASCADE_SCALE_IMAGE)

        # Process detected faces
        if not (len(faces)==0): # In any faces detected
            # Find positions of faces relative to image center
            faceCenterX = faces[:,0] + faces[:,2]/2 # left edge + half width
            faceCenterY = faces[:,1] + faces[:,3]/2 # lower edge + half height
            xOffset = faceCenterX - frameWidth*(1-desFacePos[0]);
            yOffset = faceCenterY - frameHeight*(1-desFacePos[1]);
            rOffset = np.sqrt(xOffset**2 + yOffset**2) # Radius from image center

            # Find face closest to center, calculate position error
            index_rOffsetMin = np.argmin(rOffset)
            xError = xOffset[index_rOffsetMin]
            yError = yOffset[index_rOffsetMin]

            # Update camera angle to reduce x and y error
            panAngle -= panGain*xError;
            tiltAngle -= tiltGain*yError;
            updateServoPos(panAngle, tiltAngle)

            # Draw green rectangle(s) around face(s)
            for (x, y, w, h) in faces:
                cv2.rectangle(frame, (x, y), (x+w, y+h), (0, 255, 0), 2)

        # Display the resulting frame (with or without faces)
        cv2.imshow('Video', frame)
        if cv2.waitKey(1) & 0xFF == ord('q'):
            break
```


Takk for meg!

